

The artifact I selected for this enhancement is the Python class that handles CRUD functionality for an animal shelter's database that's stored in MongoDB. This class came from an assignment done in a previous class where we did lots of database development and used MongoDB and the PyMongo library. The purpose of this artifact was to make a simple interface for the user to create, retrieve, update, and delete records from a MongoDB database, the artifact is a database access layer between the application and database. By using this class a user can perform database operations without directly creating MongoDB queries through the rest of the application.

I selected this artifact for my ePortfolio because it's a good example of my ability to work with and implement database operations through Python. Database design and implementation are important skills in Computer Science and this artifact is a showcase of my experience with MongoDB, PyMongo, and OOP techniques. The original version of this project just showed basic CRUD functionality but my enhancements add additional functionality and better reliability. Some of the enhancements I made were input validation, error handling, sorting and limiting query results, index creation, connection testing, and proper database connection cleanup. All of these enhancements can work together to demonstrate growth in database development as well as software engineering practices.

Enhancing this artifact was an opportunity for me to revisit some previous coursework and apply lots of the concepts we've learned throughout the CS program. While the original artifact worked and was successful in its use of basic CRUD functionality, when I looked at it recently it was clear there was a lot of room for improvement. While enhancing this program I was able to gain a better understanding of database design, defensive programming, and more

specifically certain MongoDB features like aggregation pipelines and indexing. I did however struggle with which improvements would be most valuable while still staying true to the original program. This enhancement is a solid example of my growth as a developer over the last few years, when I made my original project I was pretty pleased with myself and how I can work with databases now, but as I've learned more my skills have improved and now I can go back to old work and make serious meaningful improvements to my old programs.